SMART RESOURCE ALLOCATOR FOR NGOs

Data-driven volunteer coordination, intelligent resource allocation, and verified social impact

Full MVP blueprint • AI processing specification • Admin reports • Volunteer and department workflows • Routing • Data model • Flowcharts • Implementation roadmap

Product thesis: Convert scattered community data into structured needs, prioritized interventions, suitable volunteer/resource assignments, and verifiable outcomes.

Prepared for hackathon MVP planning | 23 October 2026

The key design principle

Gemma 4 should interpret messy information and draft recommendations. The application backend—not the language model—must enforce permissions, calculate operational constraints, reserve inventory, control assignments, and record the audit trail.

1. Executive summary

The product is a shared operational workspace for NGOs that receive community information from paper surveys, spreadsheets, emails, field reports, volunteer submissions, and partner groups. It should preserve the original source data, turn it into structured records, connect related reports, help administrators decide what to do, and track every approved intervention through completion and verification.

Recommended MVP crux: Import CSV/XLSX → AI-assisted structuring → admin verification → urgency scoring → volunteer/resource recommendation → approval → task execution → evidence and closure → impact summary.

Do not build this as an AI chatbot with a dashboard around it. Build a reliable workflow engine with AI embedded at specific decision-support points.

Primary roles

Role	What they do	What they should see
NGO owner / super admin	Creates organization, configures departments, roles, policies, and access.	Organization-wide metrics, audit logs, all authorized departments.
Admin / coordinator	Reviews AI reports, verifies needs, approves priorities, assignments, and resource release.	Queue of reports, task board, map, resource availability, team workload, AI rationale.
Department lead	Confirms capacity, skills, stock, and operational feasibility for their department.	Department requests, incoming referrals, assigned work, stock/capacity, handoffs.
Volunteer / field worker	Accepts work, executes tasks, updates status, submits evidence and feedback.	Only authorized tasks, instructions, contacts, allowed resources, deadlines, and relevant department channel.
Partner / community reporter (optional MVP extension)	Submits a need or shares structured information.	Their own submissions and permitted status updates; not internal sensitive records.

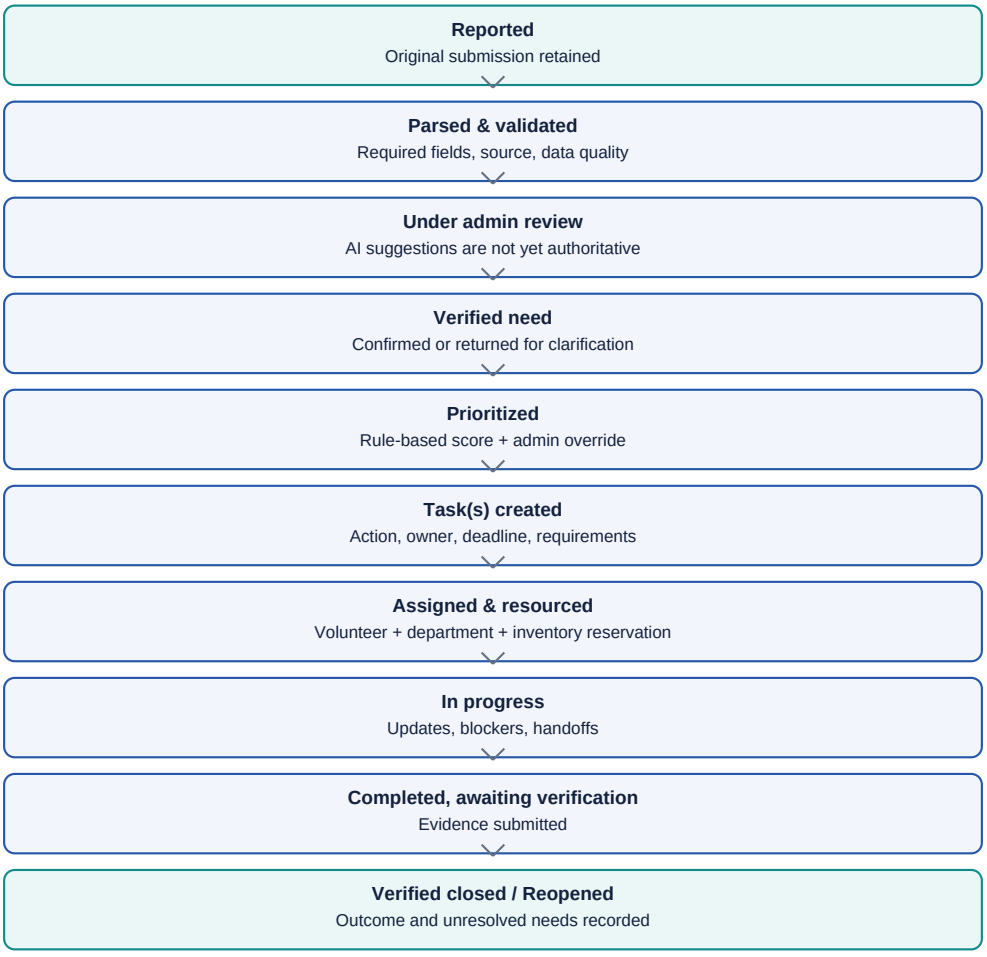
Design rule: use role-based and organization-scoped access. A volunteer should not automatically see every beneficiary record, internal admin note, or inventory location.

2. The records your system must keep separate

Do not store every incoming row as if it were a task. These entities have different meanings and different lifecycles.

Entity	Meaning	Example
Source record / report	The original observation or submission. Keep provenance.	A field worker's report; a row in an uploaded spreadsheet.
Community need / case	A reviewed underlying issue; may be linked to several reports.	Five reports about a damaged water point in the same area.
Task / intervention	A concrete action created to address a verified need.	Inspect the water point; then repair it.
Resource / inventory item	A stock item, vehicle, equipment, fund allocation, or human capacity.	20 food kits in Warehouse A; one van available Friday.
Volunteer / staff profile	Skills, verified qualifications, availability, service area, and workload.	A trained driver with a van available on Saturday.
Allocation / assignment	A traceable link between task, person, department, and resources.	Volunteer A assigned; 10 food kits reserved.
Outcome / evidence	Proof, verification, and results of work.	Delivery photo, beneficiary confirmation, admin verification.
Audit event	Who changed what, when, and why.	Admin changed priority from medium to high with rationale.

Suggested lifecycle



3. Where Gemma 4 comes into action

Gemma 4 is an open-weight model family, not the same thing as the hosted Gemini API. You can run a compatible Gemma 4 variant through a supported local inference runtime or deploy it on infrastructure you control. Exact integration depends on the model variant and runtime. Google's model documentation describes text and image input across the family, audio support for E2B, E4B, and 12B, and native function-calling capabilities. These are model capabilities—not permission to let the model execute unrestricted actions.

Recommended MVP uses

Pipeline stage	Gemma 4 can do	Application code must do
Ingestion / OCR assist	Read text extracted from reports; for supported image inputs, help interpret scanned forms, photos, charts, or document images.	Store source file securely; run OCR/parsing; preserve original; validate file size/type; redact or restrict sensitive data.
Information extraction	Return structured fields: category, department suggestion, location text, date, need, affected group, requested resources, confidence/unknowns.	Validate JSON/schema, normalize fields, resolve known department IDs, reject invalid categories.
Classification	Suggest issue category and relevant department(s), with short evidence-based explanation.	Use allowed category list; route only to configured departments; admin confirms uncertain/high-risk cases.
Duplicate/related report assist	Compare summaries and suggest likely duplicates or related cases.	Use database search, similarity thresholds, location/time rules, and human confirmation before merging.
Priority explanation	Summarize signals that may indicate urgency and identify missing information.	Compute the actual score with deterministic rules; use explicit emergency escalation; admin can override with reason.
Action plan draft	Propose next steps, checklist, skills, materials, dependencies, and risks.	Check policy, inventory, capacity, legal/safety rules, and obtain approvals.
Admin briefing	Summarize new cases, hotspots, overdue tasks, resource shortages, and decisions needed.	Calculate metrics from database queries; never let model invent counts or use unsupported facts.
Volunteer instructions	Turn an approved task into clear, localized instructions; translate/simplify if needed.	Show only authorized fields; avoid exposing beneficiary PII; require human approval for sensitive instructions.
Outcome summary	Summarize completion notes and beneficiary feedback.	Verify evidence, update inventory and status transactionally, and preserve audit history.

Do not delegate to Gemma: final priority scoring, qualification checks, stock reservation, assignment writes, route safety decisions, access control, financial approvals, emergency medical judgement, or closure verification. Use backend rules and authorized humans.

Important reliability rule

Treat every model output as an untrusted suggestion. Ask for schema-constrained JSON where the runtime supports it, validate every field, attach source references where possible, and route low-confidence or high-impact decisions to human review. Keep an explicit 'unknown / needs clarification' option so the model does not fill gaps by guessing.

4. Parameters and factors Gemma should extract

Use a stable case schema. Not every field will be available in every report; missing values should remain null and appear in a data-quality queue.

Field group	Fields to extract / derive	Validation or handling
Source & provenance	source_type, source_file, source_row/page, submitter, submission_time, source_language	Always keep original source reference; do not overwrite original text.
Problem identity	short_title, original_description, normalized_summary, report_id, related_case_candidates	Case ID is generated by backend; model may suggest links, never merge by itself.
Category & department	category, subcategory, suggested_department, related_departments	Only select from NGO-configured lists; department lead confirms referrals.
Location & time	address/locality, village/ward, district, coordinates if available, incident_date, report_date	Geocode separately; confirm ambiguous locations; protect sensitive coordinates.
Affected population	estimated_people/households, vulnerable groups if explicitly reported, access barriers	Do not infer personal sensitive traits. Record source and uncertainty.
Need & urgency signals	severity indicators, time sensitivity, immediate danger flags, service interruption, deadline	AI extracts evidence; rules calculate score; immediate danger triggers configured escalation.
Resource requirements	item, quantity, unit, required_by, special handling, funding estimate if known	Quantities and units must be normalized; unknown values stay unknown.
Human resources	skills, qualification level, headcount, language, shift, vehicle/equipment need	Mandatory qualifications are hard filters, not score bonuses.
Task proposal	recommended action, steps, estimated duration, dependencies, risk, verification method	Draft only; admin/department lead approves.
Confidence & data quality	field_confidence, missing_fields, contradictions, needs_human_review, explanation	Confidence is a model estimate, not calibrated probability unless tested.
Safety & privacy	PII present, sensitive-case flag, access restriction suggestion	Backend policy decides access; model flag alone is not security control.

Example model output (illustrative JSON)


```
{
"category": "water_sanitation",
"subcategory": "water_point_damaged",
"suggested_departments": ["field_operations", "infrastructure"],
"summary": "Reported water point is not functioning near the community centre.",
"location_text": "Village A, near community centre",
"urgency_signals": ["essential service interrupted"],
"required_skills": ["plumbing_or_water_system_maintenance"],
"resource_requirements": [{"item": "repair_kit", "quantity": null, "unit": "kit"}],
"missing_fields": ["number_of_people_affected", "date_first_observed"],
"needs_human_review": true,
"rationale": "The report describes an essential water service interruption, but scale and duration are unknown."
}
```


This is an example schema, not a promise that Gemma will always return valid JSON. Enforce schema validation in FastAPI and retry or send to manual review when validation fails.

5. Priority assessment, routing, and recommended actions

Use a two-layer design: the model extracts evidence and drafts explanations; deterministic application logic computes the operational score and routes work against configured NGO rules.

A practical priority model for MVP

Signal	Suggested weight	What it means
Severity / harm if not addressed	30%	Potential consequence of leaving the issue unresolved.
Time sensitivity	25%	Whether delay materially worsens the situation.
People affected / scale	15%	Estimated number of people or households affected.
Vulnerability / access barrier	15%	Explicitly documented barriers that increase the need for support.
Service disruption / recurrence	10%	Essential service interrupted or issue repeatedly reported.
Deadline / external dependency	5%	A known time limit or dependency that affects response.

Compute a 0–100 score only when inputs are present. Define how missing values are handled; do not silently score unknown data as zero. Keep an explicit confidence/data-completeness indicator separate from urgency.

Example priority bands (configure with NGO leadership)

Score / trigger	Queue	Expected behavior
Critical: 85–100 or explicit configured emergency trigger	Immediate escalation	Notify designated coordinator; require prompt human review. The system must not independently make medical or emergency decisions.
High: 65–84	Priority queue	Review and allocate promptly; show why it is high.
Medium: 35–64	Normal work queue	Batch or schedule based on capacity and deadlines.
Low: 0–34	Planned / monitor	Keep visible; do not lose it from reporting.

Routing logic



Department routing should use a maintained rules table, not free-form model decisions. A case can have one accountable owner department plus supporting departments. If departments disagree or no department has capacity, route to a coordinator queue.

6. Volunteer, department, and resource matching

Volunteer matching should be constraint-first, score-second. Never let a high numerical score compensate for a missing mandatory qualification, unavailable shift, or prohibited assignment.

Hard eligibility filters

- Volunteer is active and has opted in / is available for the relevant time window.
- Mandatory qualification or safeguarding checks are satisfied where required.
- Volunteer is allowed to access the organization, project, and case.
- Required language, vehicle, equipment, or accessibility needs are met if essential.
- Workload cap, shift limit, and any safety restrictions are respected.

Ranking after filtering

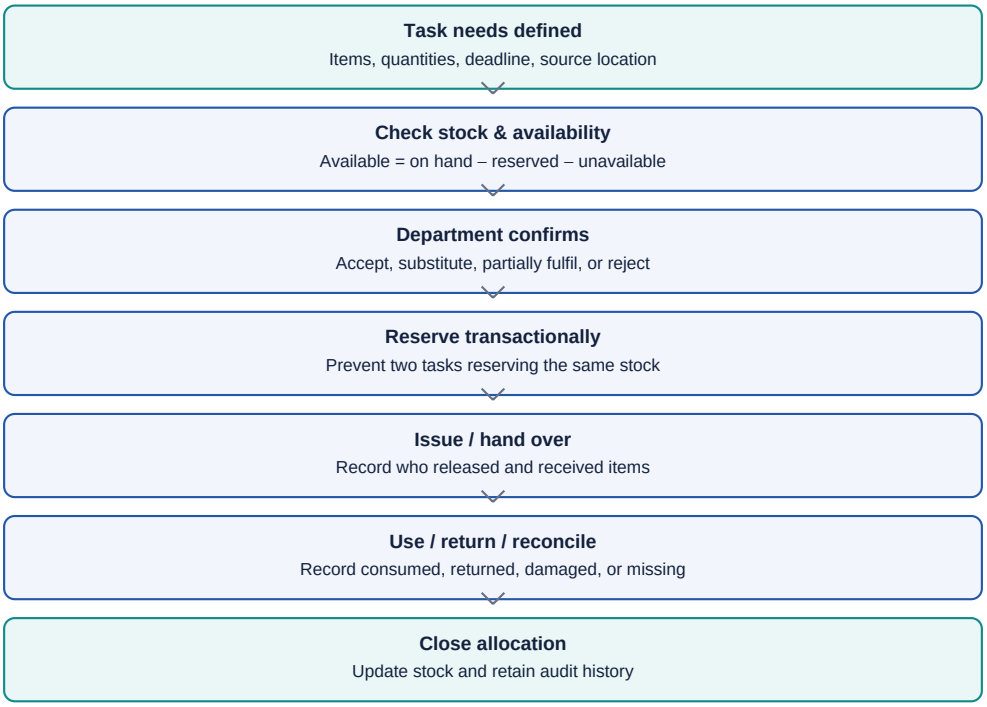
Factor	Example share	Implementation note
Required skill fit	35%	Use verified skill tags and qualification level.
Travel distance / service area	20%	Use geocoded coordinates only when available; respect privacy.
Availability / schedule fit	20%	Hard filter for unavailable; score preferred time fit.
Current workload / fairness	15%	Avoid repeatedly assigning the same high performers.
Relevant experience / continuity	10%	Use validated prior work and local familiarity.

These weights are starting points for testing, not universal truths. Evaluate match acceptance rate, completion rate, travel burden, fairness of assignments, and override frequency.

Routing is more than assigning a person

- **Organizational routing:** which department owns the case and which groups must support it.
- **Task routing:** which qualified volunteer or staff member receives the task.
- **Resource routing:** which store/department releases supplies, vehicle, equipment, or funds.
- **Physical route:** optional delivery stop ordering or map directions; do not claim optimized routes unless you implement a routing engine.
- **Escalation routing:** who is notified when a case is critical, overdue, rejected, blocked, or outside department capacity.

Resource reservation lifecycle


Task needs defined
Items, quantities, deadline, source location
Check stock & availability
Available = on hand – reserved – unavailable
Department confirms
Accept, substitute, partially fulfil, or reject
Reserve transactionally
Prevent two tasks reserving the same stock
Issue / hand over
Record who released and received items
Use / return / reconcile
Record consumed, returned, damaged, or missing
Close allocation
Update stock and retain audit history

7. What report goes to admins?

Provide two distinct reports: an **individual case intelligence report** for each new/updated need, and a **daily/weekly operations briefing** across the organization. Model-generated prose should be built from validated database facts and clearly label recommendations.

A. Individual case intelligence report

Section	What admin sees
Case summary	Plain-language description, original source link, date received, source type, submitter, case ID.
AI classification	Category/subcategory, suggested owner department, supporting departments, confidence/data-quality notes.
Need & impact signals	Location, estimated people affected, urgency evidence, known service interruption, recurring-case links.
Priority assessment	Backend score, band, contributing factors, missing inputs, emergency trigger if any, model explanation.
Recommended actions	Immediate next steps, checklist, dependencies, verification requirements, potential risks.
Resources & capacity	Required skills/headcount/items, available inventory, shortage list, departments/partners to contact.
Volunteer candidates	Eligible ranked candidates, match rationale, availability, workload, distance band, qualification check status.
Routing plan	Accountable department, support departments, referral status, escalation contact, proposed timeline.
Admin decisions	Approve/edit/reject AI suggestions, ask for clarification, merge/link reports, change priority with reason.
Audit & follow-up	Who approved, assigned, reserved stock, changed status, and when; next action and due date.

B. Daily / weekly NGO operations briefing

Section	Contents
Executive snapshot	New reports, verified needs, critical/high cases, tasks open/overdue, tasks closed, current volunteer capacity.
Needs by category/location	Counts and trend versus previous period, map hotspots, unresolved needs, reporting gaps.
Decisions needed	Cases awaiting verification, department referrals awaiting acceptance, resource releases requiring approval.
Capacity and shortages	Volunteer skills in shortage, stock below threshold, overloaded departments, unused resources nearing expiry.
Work in progress	Tasks by status, owner, deadline, blocker, last update, next responsible person.
Risk / exception list	Critical unreviewed cases, overdue assignments, repeated volunteer declines, missing evidence, inventory mismatch.
Outcome summary	People reached, resources used, response time, beneficiary feedback, unresolved/recurring needs.
Recommended admin actions	A short ranked list of suggested decisions with evidence and the responsible role.

Do not ask Gemma to invent dashboard totals. The backend calculates every count, time, stock balance, and trend. Gemma may turn those supplied facts into a concise narrative and propose decisions.

8. Admin and volunteer workspace behavior

Admin workspace

- **Inbox:** newly imported reports, failed imports, missing fields, potential duplicates, and cases needing review.
- **AI report panel:** extracted fields, priority factors, department recommendation, proposed actions, and uncertainty.
- **Needs map and queue:** filters by category, location, priority, owner department, age, and status.
- **Resource and people view:** available stock, reserved stock, volunteer availability, skills, workload, and department capacity.
- **Assignment board:** unassigned, assigned, accepted, in progress, blocked, completed, awaiting verification.
- **Decision controls:** approve/edit/reject, assign owner, request clarification, set due date, authorize resources, and record override reasons.
- **Audit and analytics:** task history, allocation history, overdue work, outcomes, and exports.

Volunteer workspace

- **My work:** tasks assigned to the volunteer, deadline, priority, status, and clear next action.
- **Task details:** purpose, safe instructions, location at the appropriate precision, required skills, estimated duration, contact protocol, and completion criteria.
- **Resources:** what can be used, quantity reserved, pickup location/contact, issue/return procedure, and responsible department.
- **Department connection:** task-specific message thread or contact link to the resource-owning department; avoid giving broad access to internal channels.
- **Progress updates:** accept/decline with reason, start, blocked, request support, upload evidence, submit completion notes.
- **Privacy by default:** reveal beneficiary contact details only when necessary for the assigned task and authorized by policy.

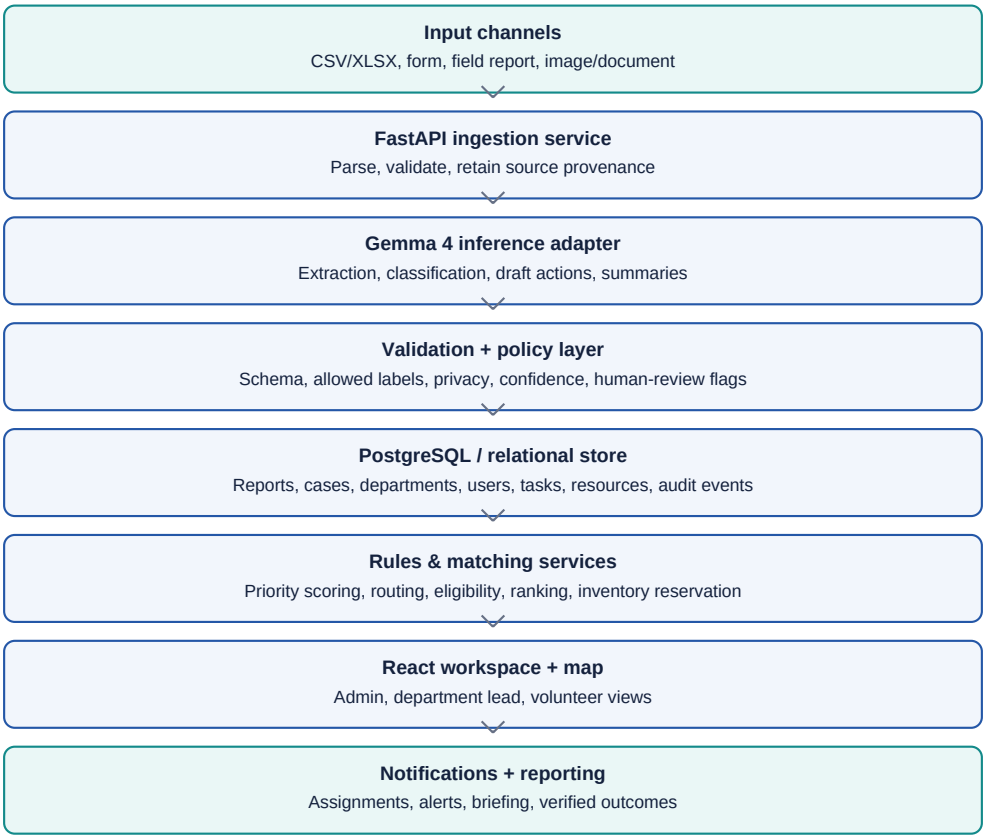
Department lead workspace

- Incoming referrals with context, requested support, deadline, and accepting/declining control.
- Capacity declaration: current staff/volunteers, stock, equipment, service area, and expected availability.
- Resource requests and release approvals with an auditable handover.
- Cross-department handoffs and escalation when the department cannot fulfil the request.

Task status transitions


Assigned
Volunteer notified
Accepted / Declined
Decline triggers reason + reassignment
In progress
Progress and blockers recorded
Completed, evidence submitted
No automatic closure
Admin verifies
Accept or return for more evidence
Verified closed / Reopened
Outcome saved; unresolved need remains visible

9. Technical architecture for the MVP



Recommended stack

Layer	MVP choice	Reason
Frontend	React + Vite; CSS/Tailwind; React Router	Fits your existing project experience and supports separate role-based workspaces.
API/backend	FastAPI + Pydantic schemas	Validation, clean API boundaries, separate services.
Database	PostgreSQL for shared deployment; SQLite acceptable for a small local demo	Relational links, transactions, audit history, concurrent users.
Auth	Firebase Authentication with backend token verification	Keep authentication separate from application authorization.
AI	Gemma 4 through a compatible inference runtime; model adapter isolates the runtime	Swap model versions without rewriting business logic.
Data import	Pandas + openpyxl / CSV parser	Spreadsheet parsing and row-level validation.
Maps	Leaflet + a geocoding provider chosen after reviewing usage terms	Visualize cases and approximate service areas.
Files	Firebase Storage or another access-controlled object store	Store source uploads and completion evidence separately from database records.

Model deployment reality check

Gemma 4 has multiple sizes and memory requirements vary by precision and runtime. Google's published estimates put Gemma 4 E2B at roughly 2.9 GB for Q4_0 weights and E4B at roughly 4.5 GB, before practical runtime/context overhead. With a 6 GB RTX 3050, a quantized small variant may be worth testing, but do not assume comfortable performance or sufficient memory for large context. Start with short inputs, limit context, benchmark on your actual machine, and keep a hosted inference option available for the demo if local latency is poor.

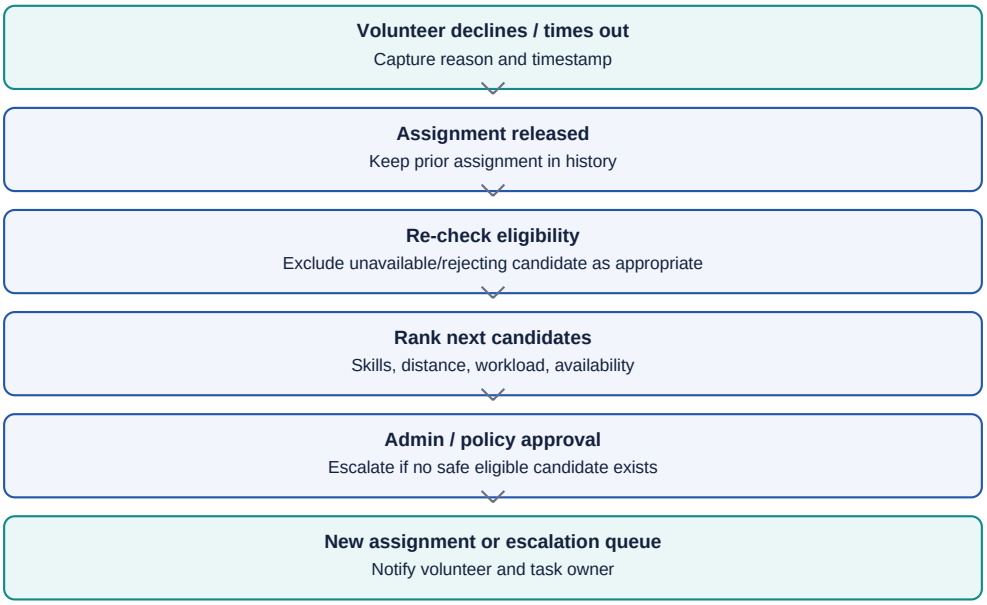
If you already have a working Gemini API integration, it is the lower-friction MVP path. If the requirement specifically says Gemma 4, integrate Gemma 4 as a separate inference service and prove that it is actually the model being called. Do not call Gemini API and describe it as Gemma 4.

10. End-to-end flowcharts

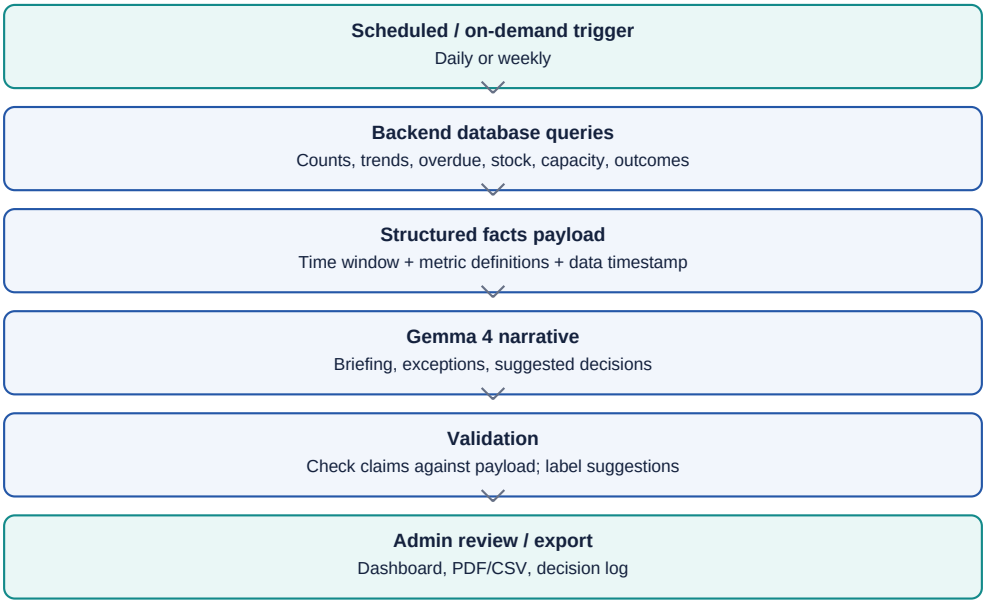
Flow A — report-to-action



Flow B — rejection and reassignment



Flow C — admin briefing generation



11. Real NGO operational controls that are easy to miss

Control	Why it matters	MVP implementation
Data provenance	Admins need to know where a claim came from.	Store source file, row/page, submitter, date, and original text.
Duplicate safeguards	Several reports can describe one incident.	Suggest related cases; human confirms merge; preserve all original reports.
Human review and appeal	AI may misclassify a case or misread urgency.	Editable suggestions, confidence/data-quality notes, reasoned overrides.
Emergency escalation	Some reports cannot wait in a normal queue.	Explicit configured triggers and human escalation path; no autonomous emergency judgement.
Privacy and safeguarding	Beneficiaries may be vulnerable; location/contact data can create risk.	Least-privilege access, data minimization, restricted fields, retention rules.
Inventory correctness	Double allocation can cause a real-world shortage.	Transactional reservation, stock movement ledger, physical reconciliation.
Volunteer safety	Some tasks require training, supervision, or safeguarding checks.	Hard qualification filters, task risk category, approved instructions, incident reporting.
Auditability	NGO accountability requires evidence of decisions and resource use.	Append-only audit events for approvals, status changes, allocations, and overrides.
Offline / low-connectivity	Field work may happen where networks are weak.	For MVP, allow draft/save-retry where feasible; mark sync state and avoid silent data loss.
No capacity / no match	The system must not invent an available volunteer.	Create a transparent shortage/escalation queue and suggest partner referral.
Fairness	Distance-only or response-rate ranking can disadvantage some volunteers/areas.	Measure workload balance, opportunity distribution, and under-reported areas.
Model failure	AI may time out, hallucinate, or return invalid structure.	Retry once, log error, allow manual processing, and never block emergency escalation.

Security boundary

The model should not receive unnecessary beneficiary names, phone numbers, exact addresses, or medical details. Send a minimized payload when possible. Never expose secrets, database credentials, or unrestricted tool access to the model. If function calling is used, expose only narrow, allowlisted backend functions; every write action must re-check authorization and business rules.

12. Suggested MVP database entities and API boundaries

Table / entity	Important fields (illustrative)
organizations	id, name, settings, active
users	id, auth_uid, organization_id, role, department_id, active
departments	id, organization_id, name, category_rules, service_area, capacity
source_records	id, source_type, file_id, row_or_page, original_text, submitted_by, received_at
cases	id, organization_id, category, summary, location, priority_score, status, owner_department_id, verified_by
case_links	case_id, source_record_id, relationship_type, confirmed_by
volunteers	user_id, skills, qualifications, availability, service_area, workload
tasks	id, case_id, title, instructions, status, priority, due_at, owner_department_id, assigned_to
task_assignments	id, task_id, volunteer_id, assigned_at, response, decline_reason, ended_at
resources	id, department_id, item, unit, on_hand, reserved, location, threshold
resource_movements	id, resource_id, task_id, quantity, movement_type, actor_id, timestamp
evidence	id, task_id, file_id, note, submitted_by, verification_status
audit_events	id, actor_id, entity_type, entity_id, action, reason, timestamp
ai_runs	id, source_or_case_id, model_id, prompt_version, output_json, validation_status, created_at

API boundaries (examples)

Endpoint group	Example purpose
POST /api/imports	Upload CSV/XLSX; return import summary and row-level errors.
POST /api/ai/analyze/{record_id}	Run extraction; store model/version and validated output.
GET /api/admin/review-queue	Return records requiring review for the current organization.
POST /api/cases/{id}/verify	Admin confirms or returns case; requires authorization.
POST /api/cases/{id}/tasks	Create an approved task from a verified case.
POST /api/tasks/{id}/recommendations	Get eligible ranked volunteer candidates.
POST /api/tasks/{id}/assign	Commit assignment after re-checking constraints.
POST /api/tasks/{id}/resources/reserve	Reserve stock transactionally.
POST /api/tasks/{id}/status	Validate allowed state transition and record audit event.
POST /api/tasks/{id}/evidence	Upload completion evidence metadata.
POST /api/tasks/{id}/verify-close	Admin verifies outcome and closes task.
GET /api/reports/operations	Return metrics from database queries for briefing generation.

13. Build order and acceptance criteria

Phase	Build	Done when
1. Foundation	Auth, organization, roles, departments, schema, audit events.	A volunteer cannot access another organization's records or admin-only endpoints.
2. Import & review	CSV/XLSX import, row errors, source provenance, review queue.	Upload produces validated records and shows rejected rows without losing original file.
3. AI assist	Gemma adapter, JSON schema, category/department suggestion, summary, missing fields.	Invalid model output is rejected; admin can edit every AI-suggested field.
4. Needs dashboard	Case queue, map, filters, priority and status.	All dashboard counts are computed from persisted data and match database queries.
5. Assignment & inventory	Eligibility filters, ranked candidates, approval, reservation ledger.	Unavailable/unqualified volunteers are excluded; stock cannot be over-reserved.
6. Volunteer workspace	Assigned work, accept/decline, status, evidence, department contact.	Volunteer sees only assigned/authorized work; decline creates a safe reassignment path.
7. Closure & reporting	Admin verification, daily briefing, outcome report.	No task closes without required evidence/approval; reports are traceable to source metrics.
8. Demo hardening	Seed data, failure states, logs, responsive UI, test scenarios.	Demo works even if AI fails by falling back to manual review.

Minimum evaluation metrics

- Import success rate and row-level error rate.
- Percentage of AI-extracted fields accepted without correction (by field, not just overall).
- Priority agreement between model-suggested rationale and admin-reviewed urgency.
- Duplicate suggestion precision and false-merge count.
- Volunteer recommendation acceptance rate and task completion rate.
- Average time from report receipt to admin review and assignment.
- Stock reservation mismatch count and overdue task rate.
- Percentage of closed tasks with verified evidence and complete resource ledger.

Demo acceptance test: Upload a needs file, volunteer file, and inventory file; inspect the AI-generated case report; correct/approve it; route it to a department; assign a qualified volunteer; reserve inventory; submit completion evidence; verify closure; and produce an admin briefing with accurate totals.

14. Sample admin case report

The following is an illustrative report generated from a fictional case. In the real system, all quantities and metrics must be sourced from verified records.

CASE SR-00042 — Water access interruption

Status: Under admin review | Suggested owner: Infrastructure / Field Operations | AI review required

Report section	Example contents
Source	Field report FR-018; received 09:20; source text retained; location described as Village A near community centre.
AI summary	Report says a local water point is not functioning. Number of affected households and duration are not specified.
Suggested category	Water & sanitation → water-point maintenance. This is a suggestion pending admin confirmation.
Priority factors	Essential service interruption reported. Scale, duration, and immediate health risks are unknown; request clarification. Backend score remains provisional until required fields are resolved.
Proposed action	Ask field worker to confirm the exact location, time first observed, affected households, and whether an alternative safe water source exists; then schedule an inspection.
Department route	Infrastructure as accountable owner; Field Operations as supporting department. Department lead must accept the referral.
Resource needs	Possible maintenance kit and qualified technician; quantities not confirmed. No stock should be reserved until the task plan is approved.
Volunteer recommendations	Show only eligible volunteers after checking verified skills, availability, workload, service area, and task risk requirements.
Admin decisions	[ ] Confirm case [ ] Request clarification [ ] Approve department [ ] Set due date [ ] Approve task plan
Audit	Every edit, approval, assignment, and inventory reservation is timestamped with actor and reason.

Admin action buttons should be explicit: Approve, Edit, Request clarification, Link to existing case, Reject with reason, Create task, Assign, Reserve resources, Escalate. The AI report itself must never silently trigger these actions.

15. Final recommendations

Keep the MVP centered on one end-to-end operational loop. The project becomes valuable when a report does not stop at classification: it becomes a verified need, an approved task, an accountable department handoff, a feasible volunteer/resource allocation, and a verified outcome.

Build now

- Structured import with provenance and error reporting.
- Gemma 4 extraction and recommendations behind a replaceable model adapter.
- Admin review queue and case report.
- Rule-based priority score with visible factors and human override.
- Configured category-to-department routing and department acceptance.
- Volunteer matching with hard eligibility filters and explainable ranking.
- Inventory ledger and transactional reservation.
- Role-specific admin, department lead, and volunteer workspaces.
- Task status, completion evidence, verification, and operational briefing.

Defer unless the MVP is already stable

- Fully autonomous assignment of high-risk tasks.
- Complex multi-stop route optimization.
- Predictive forecasting trained on historical NGO data you do not yet have.
- Cross-NGO exchange of sensitive records.
- Automatic merging of cases without human confirmation.
- Unrestricted AI agent tools that can write directly to the database.

Official Gemma 4 references

Google AI for Developers — Gemma 4 model card: https://ai.google.dev/gemma/docs/core/model_card_4

Google DeepMind — Gemma 4 overview: https://deepmind.google/models/gemma/gemma-4/

Gemma documentation — model variants and memory estimates: https://gemma.ai/models/gemma-4/

Final product definition: An AI-assisted NGO operations platform that converts fragmented reports into prioritized community cases, coordinates departments and volunteers, allocates traceable resources, and records verified outcomes. Gemma 4 provides language and document understanding; deterministic software and authorized staff retain operational control.